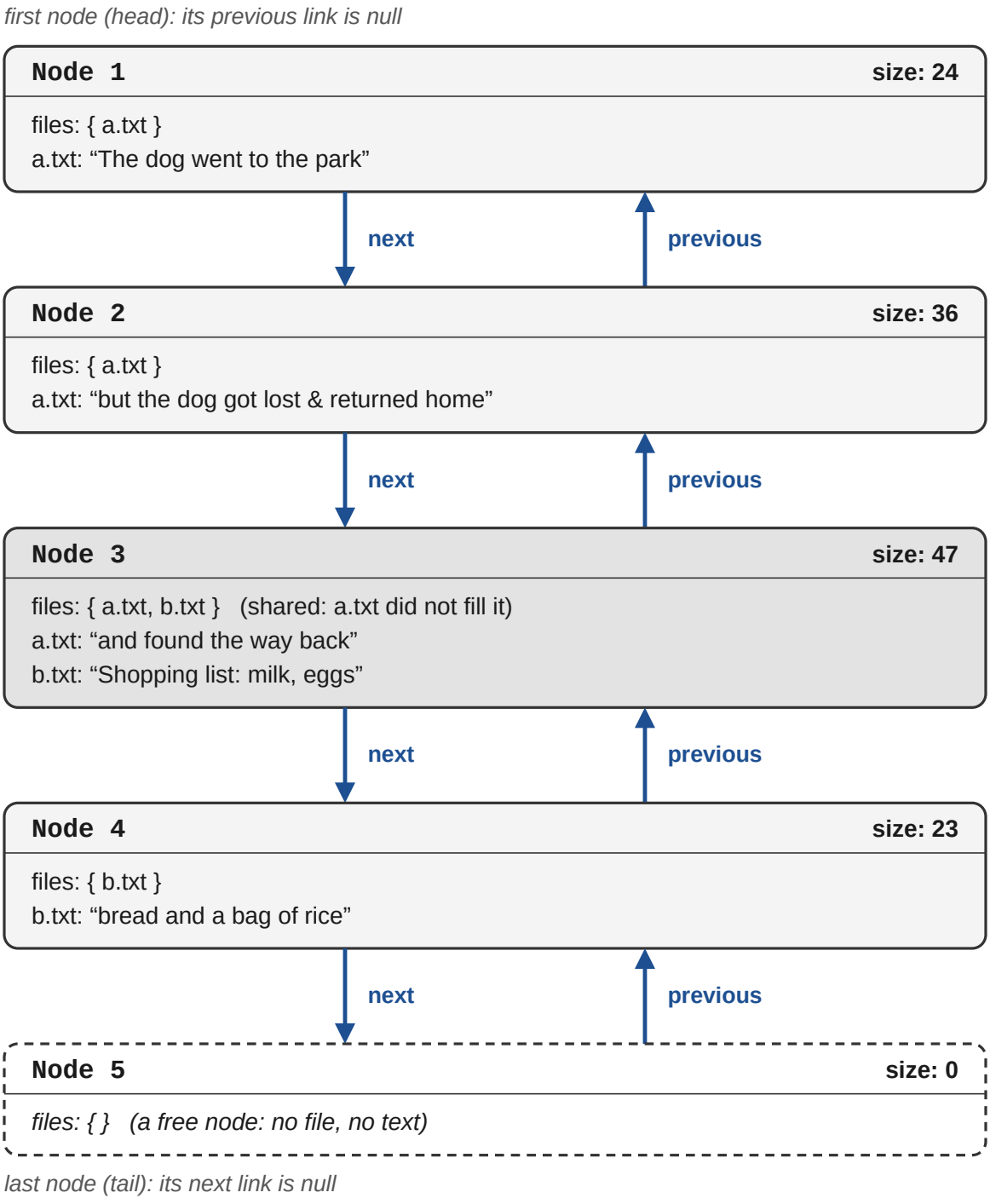
Your disk plan: a doubly linked list of nodes

Each node holds up to 1000 characters (1 KB). The example texts are shortened so they fit.



Size is the length of the text a node holds: Node 3 holds 22 + 25 = 47 characters.
A real disk of 1 MB is 1000 of these nodes, in a chain, each holding up to 1000 characters.

Node properties and the write plan

Your plan as you described it. The open points at the end are things we flagged and have not settled.

The disk

- Its size is asked for at account setup, up to 1 MB. That is 1000 KB, or 1000 nodes of up to 1 KB each.
- A body’s length divided by 1000 gives its size in KB.

Properties of a node

- **files** (the parent): a set of Files. A node that is not full can hold some of another file’s text, so a part of a KB is not lost.
- **text**: the part of each file’s body that this node holds, up to 1000 characters in total. If it is stored, it is the text of each file in the set.
- **next**: a reference to another node. It either continues the same file or starts a different one, so the disk stays connected.
- **previous**: the same, in the other direction.
- **size**: the length of the text the node holds.

Writing to a file

- Iterate over the body while setting it into the disk, one node’s worth (up to 1000 characters) at a time, for the file’s size.
- If a node is not filled, another file can use the rest of it.
- When a file already has content, find what changed, then replace or add a node for that file.

Deleting a file

- It works the same way: find the changes, and replace or remove that file’s part of each node.

Still to settle

1. **Sharing.** The assignment says a cell holds one file’s ID, sizes are whole KB, and `rm` releases every cell holding the file’s ID. Mr. Nava’s reply covered a doubly linked list instead of the `int[][]`. Your question did not mention sharing a node between files.
2. **Text and files.** If the files and the texts are two separate sets, how does each piece of text stay tied to its file? A set also has no order.
3. **Next and previous.** When a file’s pieces are not side by side, how do you get from one piece of a file to its next piece?
4. **Body to cells.** How many whole KB does a body take? For example, 456 characters.